AI Engineer Foundation System

(0 → Python → DSA → ML → AI Engineer → AI Architect)

Welcome to the **AI Engineer Foundation System**. This is a premium, execution-focused curriculum designed to take you from an absolute beginner to an AI Engineer capable of building, deploying, and architecting real-world machine learning systems.

📅 Core Roadmaps

12-Month Aggressive Roadmap (For full-time students / intensive learners)

- **Months 1-2:** Python Foundation & Basics of DSA (Parallel Track). Goal: Master syntax, OOP, and basic problem-solving.
- **Month 3:** Data Handling Mastery (NumPy, Pandas, EDA) & Intermediate DSA.
- **Months 4-5:** Machine Learning Core & Math for ML. Goal: Regression, Classification, Model Evaluation.
- **Month 6:** Advanced ML (Pipelines, Feature Engineering, Tuning) & ML Portfolio Projects.
- **Months 7-8:** Deep Learning (Neural Networks, CNNs, RNNs, NLP Basics).
- **Months 9-10:** AI Engineering & MLOps (FastAPI, Docker, Model Deployment).
- **Month 11:** Beginner AI Architecture (System Design, Scalability, Data Pipelines).
- **Month 12:** Capstone Portfolio, Resume Building, Interview Prep, & Career Kit Execution.

18–24 Month Balanced Roadmap (For working professionals / part-time learners)

- **Months 1-3:** Python Foundation (2 hrs/day).
- **Months 4-7:** DSA Parallel Track (Strings, Arrays, Trees, DP) + Python Mini Projects.
- **Months 8-9:** Data Handling Mastery.
- **Months 10-13:** Machine Learning Core & Advanced ML.
- **Months 14-17:** Deep Learning & Transformers Intro.
- **Months 18-20:** AI Engineering (Deployment, APIs, Docker).
- **Months 21-22:** AI Architecture Basics.
- **Months 23-24:** Final Portfolio, Interview Prep, and Mock Interviews.

🗂️ Premium Folder Structure & Detailed File List

Below is the exact folder architecture of this premium system.

```
AI-Engineer-Foundation-System/
├── 00_Getting_Started/
│   ├── 01_Welcome_Guide.pdf
│   ├── 02_How_to_Use_This_System.pdf
│   ├── 03_Daily_Schedule_Templates.xlsx
│   ├── 04_Study_Rules_and_Discipline.pdf
│   ├── 05_Roadmap_Overview.pdf
│   └── 06_Success_Framework.pdf
├── 01_Python_Foundation/
│   ├── Day01_to_Day05_Basics_and_Loops.pdf
│   ├── Day06_to_Day10_Functions_Lists_Dicts.pdf
│   ├── Day11_to_Day15_OOP_and_FileHandling.pdf
│   └── Python_Mini_Exercises/
```

```
├── 02_DSA_Parallel_Track/
│   ├── 01_Arrays_and_Strings/
│   ├── 02_Hashing_and_Sliding_Window/
│   ├── 03_Recursion_and_Stack_Queue/
│   ├── 04_Trees_and_Graphs/
│   ├── 05_Dynamic_Programming_Basics/
│   └── DSA_300_Curated_Problems_List.xlsx
├── 03_Data_Handling/
│   ├── 01_NumPy_Mastery.pdf
│   ├── 02_Pandas_CSV_Missing_Values.pdf
│   └── 03_Data_Cleaning_Projects/
├── 04_Machine_Learning_Core/
│   ├── 01_Regression_and_Classification.pdf
│   ├── 02_Metrics_and_Evaluation.pdf
│   └── ML_Core_Practice_Tasks/
├── 05_Advanced_ML/
│   ├── 01_Feature_Engineering.pdf
│   └── 02_Pipelines_and_Tuning.pdf
├── 06_Deep_Learning/
│   ├── 01_Neural_Networks_Basics.pdf
│   ├── 02_CNN_and_RNN.pdf
│   └── 03_Transformers_Intro.pdf
├── 07_AI_Engineer/
│   ├── 01_FastAPI_for_ML.pdf
│   ├── 02_Docker_Basics_for_ML.pdf
│   └── 03_Model_Serving_and_Deployment.pdf
├── 08_AI_Architect/
│   ├── 01_ML_System_Design.pdf
│   ├── 02_Data_Pipelines_and_Scalability.pdf
│   └── 03_Monitoring_Models.pdf
├── 09_Projects_Portfolio/
│   ├── Beginner_Projects/
│   ├── Intermediate_Projects/
│   └── Advanced_Projects/
├── 10_Problem_Sets/
│   ├── Python_100_Problems.pdf
│   ├── DSA_300_Problems.pdf
│   └── ML_Logic_Questions.pdf
├── 11_Resource_Library/
│   ├── English_Resources_Links.pdf
│   └── Hindi_Resources_Links.pdf
├── 12_Career_Kit/
│   ├── GitHub_Guide_and_README_Templates.pdf
│   ├── Resume_Project_Section_Examples.pdf
│   └── Interview_Prep_Guide.pdf
├── 13_Reference_System/
│   ├── ML_Readiness_Checklist.pdf
│   └── Cheatsheets_and_Debugging.pdf
├── 14_DSA_For_AI_ML/
│   └── Must_Do_DSA_for_AIML_Students.pdf
└── 15_Execution_System/
    ├── 01_Discipline_Framework.pdf
```

```
├── 02_Progress_Tracker.xlsx
└── 03_Weekly_Review_Template.pdf
```

📖 Module Details & Content Samples

00 GETTING STARTED

Contents:

- **Welcome Guide:** How to transition to AI engineering without getting overwhelmed.
- **Daily Schedule:** 2-hour, 4-hour, and 6-hour daily breakdown templates.
- **Study Rules:** No tutorial hell. 30% watching, 70% coding.
- **Success Framework:** Learn -> Build -> Document -> Deploy.

01 PYTHON FOUNDATION (Sample Day 6: Lists & Dictionaries)

- **Topics:** Lists, Dictionaries, Tuples, Sets, List Comprehensions.
- **Learning Objectives:** Master data storage and retrieval in memory.
- **English YouTube:** Programming with Mosh (Python Full Course - Data Structures section).
- **Hindi YouTube:** CodeWithHarry (Python One Shot - Lists & Dicts).
- **Website:** W3Schools (Python Lists & Dictionaries).
- **Time Allocation:** 2.5 Hours.
- **Practice Tasks:**
  1. Build a frequency counter for words in a string.
  2. Implement a simple key-value database for student grades.
- **Completion Criteria:** Can fetch, update, delete, and iterate through nested dictionaries without looking at syntax.
- **Common Mistakes:** Confusing list indices with dictionary keys, mutating lists while iterating over them.

02 DSA PARALLEL TRACK (Sample: Sliding Window)

- **Why it matters for AI/ML:** Used in time-series data parsing, NLP token windowing, and optimizing consecutive data sweeps.
- **English Resource:** NeetCode (Sliding Window Playlist).
- **Hindi Resource:** Krish Naik (DSA Logic building) / CodeWithHarry.
- **Questions:**
  - *Q1:* Maximum Sum Subarray of Size K (Easy)
  - *Q2:* Longest Substring Without Repeating Characters (Medium)
- **Hints:** Expand right pointer, shrink left pointer when condition breaks.
- **Complexity:** O(N) Time, O(1) Space.

03 DATA HANDLING

- **Core Focus:** NumPy (Vectorization), Pandas (DataFrames), CSV handling.
- **Projects:** "Simple Data Cleaner" - Load a messy CSV, handle NaNs using mean imputation, drop duplicates, and export to a clean CSV.

04 MACHINE LEARNING CORE

- **Topics:** Linear Regression, Logistic Regression, Decision Trees, Precision, Recall, F1-Score.
- **Resources:** Andrew Ng (Machine Learning Specialization - Coursera/YouTube).
- **Project:** "House Price Predictor" - Use Scikit-Learn to build, train, and evaluate a regression model.

05 ADVANCED ML

- **Topics:** Feature Engineering, Pipelines (Scikit-Learn `Pipeline`), Cross-Validation, GridSearchCV.
- **Goal:** Write production-ready, leak-proof training code.

06 DEEP LEARNING

- **Topics:** Forward/Backward propagation, CNNs for Vision, RNNs for Sequence data, Transformers Intro.
- **Resources:** freeCodeCamp (PyTorch for Deep Learning Full Course).

07 AI ENGINEER

- **Topics:** FastAPI, Docker basics.
- **Project:** "ML API" - Wrap the House Price Predictor in a FastAPI backend and Dockerize it.

08 AI ARCHITECT

- **Topics:** ML System Design, Scalability, Data Pipelines, Monitoring (Data Drift).
- **Focus:** Beginner-friendly introduction to designing systems that serve millions of inferences.

📂 09 PROJECTS PORTFOLIO

Beginner

- **Marks Analyzer:** Python CLI tool to calculate class averages and top performers.
- **Expense Tracker:** CLI/File-based tracker using dictionaries and JSON.
- **CSV Cleaner:** Pandas script to format messy Excel dumps.

Intermediate

- **Recommendation System:** Collaborative filtering for movie recommendations.
- **Spam Classifier:** NLP text classification using Naive Bayes & TF-IDF.
- **House Price Predictor:** Scikit-learn regression model.

Advanced

- **End-to-End Deployed System:** An NLP sentiment analysis model exposed via FastAPI, containerized with Docker, and hosted on Render/AWS.
  - **Dataset:** Kaggle IMDB Reviews.
  - **Expected Output:** A working public URL where users post text and get sentiment scores.
  - **GitHub Structure:** Includes `app/` , `Dockerfile` , `requirements.txt` , `model/` , and `README.md` .

📝 10 PROBLEM SETS (Excerpts)

- **Python:** 100 problems focused on logic and data structures.
- **DSA:** 300 problems (LeetCode/NeetCode curated list matching AI needs).
- **ML Logic:** e.g., "Why is accuracy a bad metric for highly imbalanced datasets like fraud detection?"

📚 11 RESOURCE LIBRARY

Trusted Sources Used:

- **English:** freeCodeCamp, Andrew Ng, NeetCode, W3Schools, Kaggle Learn.

- **Hindi:** CodeWithHarry, Krish Naik.

💼 12 CAREER KIT

- **GitHub Guide:** How to write a `README.md` that recruiters actually read.
- **Resume Section:** Action-driven bullet points (e.g., "Deployed a FastAPI inference endpoint handling 100+ requests/sec using Docker").
- **Interview Prep:** How to explain model selection and trade-offs.

📑 13 REFERENCE SYSTEM

- **ML Readiness Checklist:** Can you explain Bias-Variance Tradeoff? Can you write a Scikit-Learn pipeline?
- **Debugging Checklist:** Are the tensors matching shape? Is the learning rate too high?

🧠 14 DSA FOR AI/ML

- **Must-Do Curated List:** Focus on Matrix manipulations, Graph traversals (for Neural Net understanding), and Optimization (Greedy/DP for loss functions).

⚙️ 15 EXECUTION SYSTEM

1. **Why Students Fail:** Tutorial hell, skipping math, not building end-to-end.
2. **Discipline Framework:** Code 1 hour a day, no exceptions.
3. **Stuck Protocol:** 15-minute rule -> Google Error -> Read Docs -> Ask ChatGPT -> Move on temporarily.
4. **Weekly Review Template:** What did I learn? What project did I push to GitHub?

This document serves as the master blueprint. Use it to structure your learning, validate your progress, and build a career-ready portfolio.